Education Expansion as a Policy Tool for Reductions in Infant and Maternal Mortality*

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February 19, 2024

Abstract

We investigate the association of women's education with maternal and child mortality. Using a large panel of data consisting of 185 countries over 30 years, we find that moving an additional 1 percentage point of women into primary education reduces rates of maternal death by 4.6%, and reduces the rate of infant deaths by 1.9%. These relationships are non-linear, with secondary education having a smaller impact than primary, and tertiary education having no impact on average. The impacts of primary education are larger in South and Central Asia than elsewhere, but present in all regions other than Europe and North America. Our estimates reveal that a one standard deviation increase in the percent of women with primary education has larger impacts on mortality than a one SF increase in GDP p.c. or public health provision.

JEL codes: H52, I15, I18, J13.

*We are grateful to Axel Schoerner Emillon for outstanding research assistance and to Daniel Halim who acted as a discussant of this work at the 2023 ADBI Annual Conference, providing really insightful feedback. Sonia Bhalotra acknowledges partial support for her time from the MiSoC (Essex) and CAGE (Warwick) ESRC research centre grants.

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1 Introduction

Across the world, women's education has risen rapidly, and more rapidly than that of men in the last three decades. In this paper we investigate the extent to which this has driven the unprecedented decline in maternal mortality over this period, and the similarly notable decline in infant mortality.[1] The main analysis uses a panel of data for 185 countries during 1990-2020, but we also provide results for each of the six broad regions of the world to allow for varying baseline rates of both women's education and mortality, and for different institutions and norms that might weaken or strengthen the relationships of interest. We primarily analyse changes in the educational attainment of women of fertile age, but we then allow for heterogeneity in the relationship by age, across the full age range, as this throws light on the underlying mechanisms – as well as on the cohort-lag that any policy innovation will entail. We consistently examine the impacts of expanding the share of women in primary, secondary and tertiary education. This is relevant to identifying policy targets and to understanding the nature of the female-education input in the underlying survival production function. Acknowledging that individuals do not necessarily complete education in any one block, we also provide results using a continuous measure, which is the mean years of education of women in a country.

The Global Gender Gap Report of 2020 reveals that, in the year 2020, 88% of women worldwide had completed primary education, compared to 91% of men. The share of women completing secondary education, at 66%, had converged to equal that of men, and the share of women with tertiary (university) education was higher than the share for men- 41% compared with 36%. This represents a considerable advance but women still lag men in primary school attainment and, notably, only in primary school attainment. At risk of stating the obvious, this suggests that the real barrier for women lies in accessing basic education– which is of course a necessary condition for further education. Once they start, women's attainment progresses as well as or better than men on average. However, some women are "left out" and these women are "at the bottom of the pile" with no education. The average figure of 12% of women worldwide without an education (relative to 9%) of men represents around 360 million women aged over 15 years. As discussed, we will investigate the extent to which primary as opposed to further education can influence maternal and child mortality reduction.

[1] From 2000 to 2020, the global maternal mortality ratio (MMR) declined by 34 per cent – from 339 deaths to 223 deaths per 100,000 live births, according to UN inter-agency estimates. In the same two decades, the global infant mortality rate declined from 55 to 28 deaths per 1000 live births.

The current policy recommendation for addressing global maternal mortality, framed by the World Health Organization (WHO), is the expansion of reproductive health service coverage in low and lower middle income countries. This reflects the scientific understanding that the main proximate causes of maternal death are pregnancy complications such as pre-eclampsia or bleeding, infections and unsafe abortion.[2] These risks can all be managed by skilled professionals being present at birth. Maternal death is associated with high fertility, fertility at young ages and unsafe approaches to abortion, all of which are responsive to availability of contraception. Thus the recommendations for supply of public health services are spot on. However, there are potential frictions in women taking up contraception, safe abortion and skilled attendance at birth. For instance, Mohanan et al. (2014) found that a voucher programme introduced by the government in the Indian state of Gujarat to encourage institutional delivery experienced poor uptake. While this may reflect the poor quality of public services or transport costs, it is also possible that it reflects information frictions and that information frictions are reduced by education. It is notable that discussion of maternal mortality by national and international policymakers seldom indicates women's education as a cause, or a solution. The academic literature has similarly very little to say about the relationship between maternal mortality and women's education.

On the other hand, there is a lively literature in economics that documents a positive correlation between education and other indicators of health. Education has been shown to causally increase engagement in healthier behaviours including avoidance of smoking and drinking, seeking of prenatal care, adoption of new drugs or adherence to treatment for disease (Goldman and Smith, 2010; Goldman and Lakdawalla, 2001; Currie and Moretti, 2003; Lichtenberg and Lleras-Muney, 2005; Cutler and Lleras-Muney, 2010; Glied and Lleras-Muney, 2008; Jensen and Lleras-Muney, 2012). One explanation of why education increases healthy behaviours is that it makes it easier to acquire and process new information (Rosenzweig, 1995; Rosenzweig and Schultz, 1989; Cutler and Lleras-Muney, 2010). Information processing in our setting may pertain to the benefits of immunizations, new technology including contraceptive methods, new medicines, the rules around use of a voucher, or a new institutional setting for birth. The payoff to education tends to be higher in circumstances where learning is important, for instance, the circumstances of structural and technological change (Foster and Rosenzweig, 2004), which characterise developing countries in the post-war period; also see (Cutler et al., 2006), who highlight massive technological change in the

[2] We provide a slightly more extensive discussion of the medical literature related to maternal mortality and death in and surrounding childbirth in online appendix A.

public health sphere. A related but potentially distinct mechanism is that education increases the efficiency of a given set of health inputs. For instance, if the input is medication, education may increase adherence to or the continuation of treatment regimes. Another potential mechanism is that education modifies time preference, encouraging investment in future outcomes. A third is women's education tends to lower their *desired* fertility (Endale Kebede and Goujon, 2022). A fourth mechanism is that education improves women's bargaining power in the household and that this allows them the autonomy to look after their health and greater reproductive control, which manifests as translating their desired fertility into their actual fertility (for instance, by using contraception). Men tend to desire higher fertility than women and to use coercive control to achieve this, see Bhalotra and Rawlings. (2023). Yet another mechanism is that education raises earnings and labour market attachment (see, eg. González et al. (2023)), and this facilitates acquisition of money-intensive inputs to health and survival, albeit it may hamper investment in time-intensive inputs if educated people work longer hours. Even if this is a likely channel, it is unlikely to be the only channel because (as we shall also confirm) education tends to impact health conditional upon income (Lleras-Muney, 2005; Cutler and Lleras-Muney, 2010).

We have so far discussed mechanisms relevant to maternal mortality. In fact, we expect that a woman's education is protective not only of her own health but also of the health of her children. The likely mechanisms by which maternal education impacts infant mortality are similar, though not identical as the etiology of maternal and infant mortality is different.Infant mortality is defined as the death of the newborn child in the 12 months following birth. Maternal mortality is defined to occur in the 42 days following the termination of pregnancy and postpartum infections can be harmful to the health of the mother without being of consequence for the health of the child. While, as discussed, avoidance of maternal mortality is closely tied to the coverage and quality of reproductive health services, infant mortality is more dependent upon public health provision of immunization and other postnatal care (such as encouragement of breastfeeding), sanitation, clean water and the avoidance and management of childhood infectious diseases.

We rely on panel data techniques to seek identification, acknowledging that it leaves open threats to causal identification, primarily that the estimated impacts of women's education on the mortality outcomes reflect omitted country-year varying factors that are correlated with both women's education and mortality. We control for the likely confounders and we argue that the "stock" nature of women's education mitigates bias. We find large and consistently significant

impacts of women's education in reducing both maternal and infant mortality. Indeed, depending upon the specification, the (standardized) impacts of primary education on maternal mortality are equal to or larger than the impacts of skilled birth attendance, an instrument that the WHO currently focuses upon in devising programs to address maternal mortality.

The relationships are non-linear- the largest survival gains derive from primary education. Secondary and, in cases, tertiary education deliver significant additional gains, but there are diminishing returns. Using continuous years of education, we see that the marginal effect of education on MMR plateaus after 6 years of schooling (corresponding to primary) whereas the marginal effect on IMR only plateaus after 9 years of schooling (suggesting a return to secondary education). As discussed earlier, in the analysis period, women lagged men in achieving primary education, while overtaking men in tertiary education completion rates. Our results underline an important negative externality from failing to fill the gap at the bottom. Put differently, our results highlight that universal primary education is a potential policy tool for maternal and child mortality reduction.

The size of the impact of women's education is monotonically declining in women's age. This implies that education is most beneficial for younger women in terms of their own and their child's mortality risk. To understand the mechanisms driving the observed impacts of women's education, we investigate conditional descriptive associations of education with indicators of fertility, and indicators of reproductive health coverage. We find a significant association in the expected direction of education with professional attendance at birth, contraceptive use and fertility, with particularly large impacts on teenage fertility. We see some sharp differences in the relationships of interest across continents, with women's education having larger impacts on both maternal and infant mortality in South and Central Asia than elsewhere. The only region where women's education has no impact on these outcomes is Europe and North America, in line with primary education being essentially universal in this region and also with the low hanging fruit for maternal mortality reduction having been plucked.

2 Data

Measures of educational attainment are collated from the cross–country dataset compiled by Lutz et al. (2018) from the European Commission and Wittgenstein Centre (hereafter 'Wittgenstein data'), which provides a measure of total years of schooling, along with the proportion of the population with each level of primary, secondary, and tertiary education or with no formal educa-

tion. The Wittgenstein data allow us to observe the evolution of national attainment figures over time in five-yearly age groups, both for the entire population, and for the subgroup of females. The age-specific nature of this dataset allows for us to limit our analysis to individuals of fertile age, which for simplicity we define as between the ages of 15 and 39.[3] In principal models we focus exclusively on the impacts of educational expansions among fertile aged women, though in certain analyses we also document impacts of educational expansions at other ages. Specifically, in these analyses we consider the proportion of individuals with no education, primary education, secondary, and tertiary education reported in Wittgenstein data for each age group in 5 year age groups between 20-24 years, 25-29 years, …, 70-74 years, and at ages $\geq$75.

These education data are merged by country and year with information on maternal mortality and infant mortality. As principal measures we focus on, firstly, the maternal mortality ratio, defined as the number of maternal deaths per 100,000 live births. These data are provided by the World Health Organization (WHO, 2012, 2023), generated by a multi-agency UN workgroup known as the Maternal Mortality Estimation Inter-Agency Group (MMEIG). The maternal mortality ratio indicator is calculated by the MMEIG using figures collected in household surveys and administrative records for maternal mortality and live births. Maternal deaths are classified (ICD 10) as those mothers who die from any cause "related to or aggravated by pregnancy or its management (excluding accidental or incidental causes) during pregnancy and childbirth or within 42 days of termination of pregnancy…".[4] Given concerns of underreporting by member states and the difficulty in compiling accurate measures of maternal mortality, a Bayesian technique is used to consider as wide a possible range of vital statistics and survey datasets (Alkema et al., 2015). While we focus on the maternal mortality ratio as a principal measure of maternal health given that it calculates the adjusted risk of deaths per live birth, in supplementary analyses we also examine the lifetime risk of maternal death, which is adjusted to recognise that the total risk of death depends upon aggregate fertility rates. Specifically, the lifetime risk for a country c at time period t is calculated from the maternal mortality ratio (hereafter MMR) as $1-(1-MMR_{ct})^{TFR_{ct}}$, where TFR is the total fertility rate.[5]

Secondly, we access information on infant mortality rates by country and year which are gen-

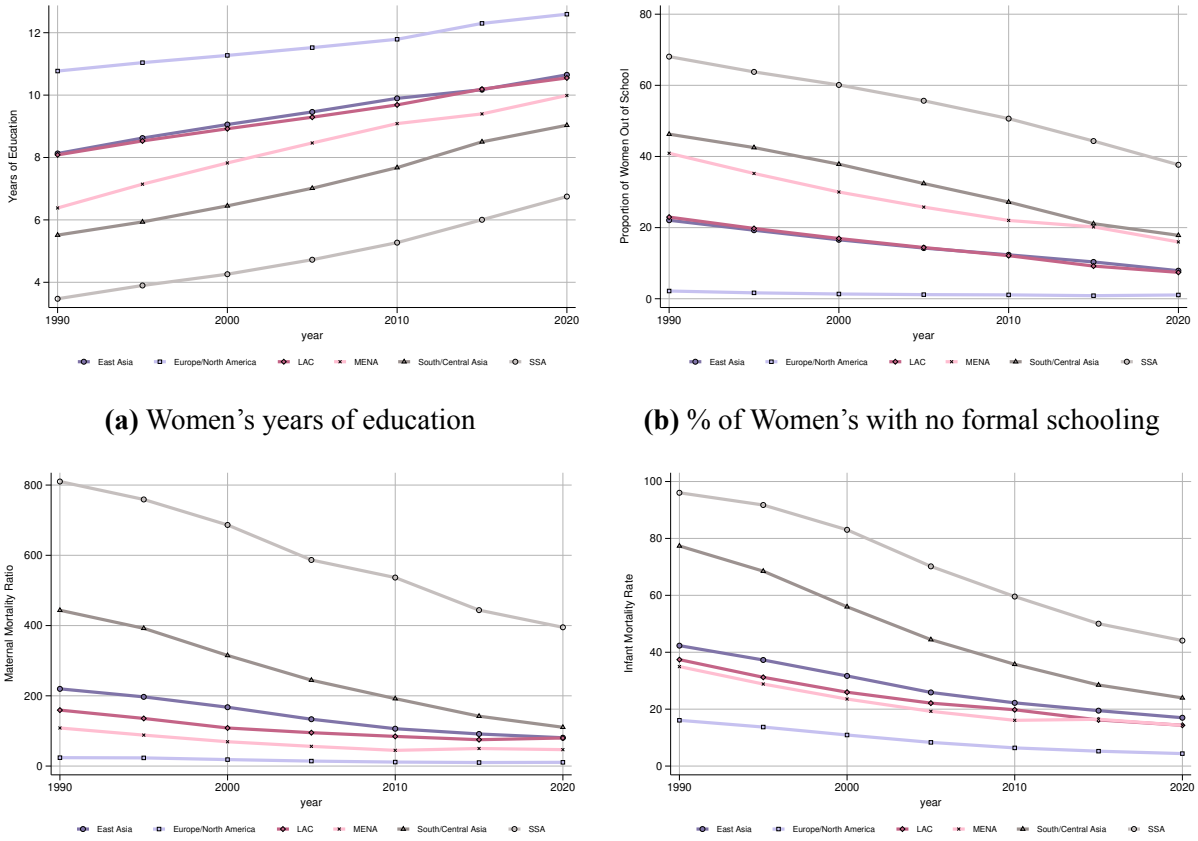
[3] Similar estimates are run for the age group 15-49. The qualitative implications of the estimates do not change.

[4] A full description is provided by The World Health Organization at http://apps.who.int/gho/indicatorregistry/App_Main/view_indicator.aspx?iid=26

[5] It is worth noting that, in practice, this quantity is quite similar to the calculation $MMR \times \frac{\text{live births}}{100{,}000}$. In our data, the relative difference between lifetime risk and this quantity is at most 5% and on average approximately 0.5%.

erated by the UN Inter-agency Group for Child Mortality Estimation. These estimates exist yearly from 1960 until 2021, and we use data with an identical periodicity as principal education measures, namely quinquennially between 1990 and 2020. These data similar combine a broad range of demographic surveys, administrative, and censal sources to estimate infant mortality by country and year based on a Bayesian posterior (refer to Alkema and New (2014); Alexander and Alkema (2018) for discussion).

Figure 1: Regional Trends: Mortality and Educational Measure



(a) Women's years of education

(b) % of Women's with no formal schooling

(c) Maternal mortality ratio

(d) Infant mortality ratio

Notes to Figure 1: Each panel presents temporal trends in key variables over the period under study (1990-2020) for 6 broad graphic regions (East Asia and the Pacific, Europe and North America, Latin America and the Caribbean, the Middle East and North Africa, South and Central Asia, and Sub-Saharan Africa.

Wittgenstein data and infant and maternal mortality measures are combined in a panel consisting of a maximum of 185 countries with observations over a period of maximum 30 years (1990-2020). Both maternal mortality and educational measures are based on measures starting in 1990, and

measured every 5 years up until 2020. Infant mortality is similarly sub-set to every fifth year. It is important to note that the Wittgenstein data for 2020 is a projection, with the last observed year being 2015. In practice, the 185 countries we observe at some point in our panel do not have observations for all years; a small number of countries do not have information on maternal mortality in the initial years under study, and certain countries like South Sudan did not exist in the initial years studied.

Descriptive trends of all key measures are provided in Figure 1. These present regional averages for 6 geographic regions for mean years of education for women of fertile age (panel (a)), the proportion of women with no formal schooling (panel (b)), the maternal mortality ratio (panel (c)), and the infant mortality ratio (panel (d)). In each case, impressive movements are observed, though both at the baseline (1990) and end line (2020) substantial regional variation is observed in all outcomes. For example, in sub-Saharan Africa, mean female years of education (among 15-39 year olds) increased from less than 4 years in 1990 to nearly 7 years in 2020. And the maternal mortality ratio in this group nearly halved, from 800 in 1990 to 400 in 2020. However, despite these large gains, maternal mortality is around *thirty times* higher in Sub-Saharan African countries than in Europe and North America in 2020, a figure which is slightly *worse* than the difference in 1990. Across all regions considered, there is a clear cross sectional relationship between women's education and maternal and infant mortality. Below, we examine whether this relationship is maintained even when conditioning on aggregate country and year specific factors, as well as other concomitant changes which occurred in the 30 years under study, such as economic growth and expansions in health care access.

To these main sources of information we add a number of covariates which we consider as potentially relevant confounding variables in our analysis of interest. These measures are generally available by country and year, and are systematised as part of the World Bank databank. In particular, these measures are: GDP per capita (PPP adjusted, in constant 2017 USD), immunization rates[6]—which we consider an imperfect proxy for health expenditures and health system capacity—and fertility and teenage fertility, and the proportion of births attended by skilled health staff. For all variables, measures are harmonised and cover the widest group of countries and time periods possible over the period of 1990-2020. In a small number of cases, measures for these variables will not be available for 1990, 1995, …, 2020, but will be available in adjacent years, and

[6]Specifically, immunisation rates measuring the proportion of children inoculated against diphtheria, pertussis (whooping cough), and tetanus, which is known as the DPT vaccine.

in these cases to avoid elevated rates of missing information in panel analysis, we impute values from adjacent years.

Table 1: Summary Statistics

	Mean	SD	Min.	Max.	Obs.
Panel A: All country× year observations					
Maternal mortality ratio (per 100,000 births)	220.72	318.89	1.10	1768.00	1238
Lifetime risk of maternal death (%)	1.12	1.98	0.00	12.00	1238
Infant Mortality Rate (per 1,000 live births)	34.01	32.11	1.70	175.80	1231
Years of schooling (Female)	8.55	3.46	0.10	14.14	1238
Primary education (% Female)	14.43	12.07	0.00	61.75	1238
Secondary education (% Female)	47.61	23.90	0.02	91.04	1238
Tertiary education (% Female)	14.04	12.54	0.00	65.09	1238
No completed education (% Female)	23.91	28.02	0.00	98.62	1238
GDP per capita (PPP, $1000 USD)	16.67	18.83	0.46	114.34	1167
Immunization (DPT) (% coverage)	83.96	16.83	17.00	99.00	1215
Fertility per woman	3.22	1.71	0.84	8.61	1238
Teen fertility (births per 1,000 teens)	61.83	47.52	1.17	210.38	1238
Female age of marriage	24.04	3.58	15.00	34.00	797
Attended births (%)	81.39	24.83	5.00	100.00	981
Contraception use (%)	46.62	22.79	1.70	87.70	718
Panel B: Baseline (1990) values					
Maternal mortality ratio (per 100,000 births)	312.15	407.25	5.12	1553.00	173
Lifetime risk of maternal death (%)	1.80	2.67	0.01	10.57	173
Infant Mortality Rate (per 1,000 live births)	50.03	40.58	4.60	175.80	172
Years of schooling (Female)	7.25	3.61	0.10	12.63	173
Primary education (% Female)	16.43	12.78	0.00	57.35	173
Secondary education (% Female)	41.90	27.61	0.02	90.12	173
Tertiary education (% Female)	9.18	9.16	0.00	45.55	173
No completed education (% Female)	32.49	32.51	0.00	98.62	173
GDP per capita (PPP, $1000 USD)	12.76	15.52	0.46	109.69	148
Immunization (DPT) (% coverage)	77.56	19.32	17.00	99.00	163
Fertility per woman	4.03	1.91	1.33	8.61	173
Teen fertility (per 1,000 teens)	78.49	50.28	3.48	210.38	173
Female age of marriage	23.28	3.08	17.05	33.70	125
Attended births (%)	77.73	26.12	7.40	100.00	94
Contraception use (%)	41.26	23.53	1.70	81.20	77

Notes: Each observation consists of a country×year pair. Summary statistics are presented for the full sample period of 1990-2020 in Panel A, and only for baseline values (1990) in panel B. Units of measurement are indicated in variable names.

Summary statistics for all variables are displayed in Table 1. Statistics for the entire sample

are provided in Panel A, while values at baseline (year 1990) are presented in panel B. At various times throughout the paper, we will express results in terms of baseline means drawn from Panel B. Further temporal trends of these measures, along with distributions or variables at each point in time, are presented in Appendix Figure A1 (mortality), and Appendix Figure A2 (education). Over time, these figures show clear declines in both median and maximum mortality values and for the proportion of women with no education or only primary education, with corresponding increases in median and maximum values of secondary education and tertiary education coverage.

3 Methodology

We examine the relationship between mortality and education in the following panel fixed effect specification:

$$\text{Mortality}_{ct} = \beta_0 + \beta_1 \text{Primary}_{ct} + \beta_2 \text{Secondary}_{ct} + \beta_3 \text{Tertiary}_{ct} + X'_{ct}\Gamma + \mu_c + \phi_t + \varepsilon_{ct}. \quad (1)$$

Mortality_{ct} refers to mortality or infant mortality rates for country c in year t, and Primary, Secondary, and Tertiary refer to contemporaneous measures of population-level means for the proportion of fertile-aged women with each particular education level in this country and year. We will progressively include a series of time-varying country-level controls such as GDP per capita measures of health system capacity, and fertility measures, indicated above X_{ct}.

Given that we have both temporal and regional variation in the variables of interest, we employ a fixed-effects methodology which allows for us to control for all time-fixed country-specific factors which may confound the analysis in (1). The ability to eliminate these time invariant factors is particularly important in this analysis given that many unobserved factors, such as the institutional and political aspects of a given country, are likely to be correlated with both maternal or infant mortality and educational attainment. Whilst some of these variables are not strictly "time-invariant", if they evolve sufficiently slowly that they are effectively fixed for a given country in the 30 year period we examine, a fixed-effect estimator will largely control for them. We cluster standard errors at the level of the country.

Bias in estimates will exist to the degree that any relevant factors which are not fixed by country or by time are omitted from (1). Here, as standard, relevance refers to factors which are related to both education, and to health, but which are not themselves mediating factors in this relationship of

interest. For example, factors such as economic growth drive both expansions in public investment in education and health. And factors such as increasing aid receipt may be conditional on investments in health and education. While covariates allow us to capture certain observable factors, it is infeasible to capture all such relevant factors because there are many confounding factors, and some of these confounding factors are inherently unmeasurable. Hoever it is important to note that the timing of accrual of education is relevant in understanding any remaining bias. In particular, education measures a stock, rather than a flow at a particular moment of time t. This is relevant, given that any decisions related to education by fertile aged women will have been taken years in the past. For example, the share of fertile-age women (who have a median age of 27 in our data) with primary education in year 2000 will have made the decision to acquire that education on average 15-20 years prior. When we model mortality in year 2000 as a function of education stocks in year 2000, the education stock is in fact pre-determined with a lag of 10-15 years, making it unlikely that it is correlated with shocks to the outcome (mortality) in year 2000. Nevertheless, because we cannot rule out all possible confounders, we will refer to any estimates as conditional correlations, recognising that estimates will include any causal relationship between education and health, as well as joint unobservable factors.

Nevertheless, we will consider the informativeness of these coefficients as potentially reflecting true causal relationships by examining the evolution of coefficients on education as a progressively more complete set of controls is included. While it is difficult to capture all relevant factors; and indeed, some such as women's rights and women empowerment are likely relevant and multi-dimensional in nature (Bhalotra et al., 2023) the stability of coefficients across the sequence of controls is informative of the existence of causal relationships. If the inclusion of theoretically highly relevant controls does little to mute effects of education on mortality, this will suggest that unobservables would need to be substantially *more* important than observed factors for reported effects to be driven purely by omitted other factors (Altonji et al., 2005; Oster, 2019).

We estimate (1) for a number of different subgroups. Whilst we primarily focus on women of reproductive age, our data allows us to run a similar analysis for various age subgroups created both within and outside of the fertile age range. We also segregate countries by region and present group-specific estimates of the above specification. Regional estimates in particular are useful in informing which factors may explain larger translations of education into mortality effects

Since our primary interest in this paper is in the benefits flowing from educating women, we ad-

ditionally consider the extent to which within-country improvements in women's education move some of the proximate determinants of MMR and IMR. We consider fertility-related variables (age at marriage, control over reproductive behaviour, fertility, teen fertility), an indicator of reproductive health coverage (share of births attended by a skilled professional) and the share of women using (modern) contraception, variation in which can reflect either or both of fertility preferences and reproductive health coverage. We regress these variables, one at a time, on education conditional upon country and year fixed effects:

$$W_{ct} = \gamma_0 + \gamma_1 \text{Primary}_{ct} + \gamma_2 \text{Secondary}_{ct} + \gamma_3 \text{Tertiary}_{ct} + X'_{ct}\xi + \mu_c + \phi_t + \eta_{ct}. \tag{2}$$

All definitions in (2) follows that in (1), however here, potential mediators are included on the left, to provide descriptive measures of how educational expansions at different margins (and in different samples) drives potential mediators, allowing us to consider whether movements in these variables correspond to movements in mortality. This will be illustrative particularly in regional analyses to potentially shed light on how and why educational returns vary across setting.

4 Results

When we present estimates for maternal mortality these will always be per 100,000 live births, while estimates for infant mortality will be per 1,000 live births. This follows conventions in the literature. The population level rates will occasionally be abbreviated as MMR and IMR respectively.

4.1 Descriptive associations within and across countries over time

Figure 2 presents a descriptive plot of changes in mortality against changes in education over time at the country level, where the change is a long difference, taken over 1990-2020. Each panel thus shows, for a given country, how maternal (top panel) and infant mortality rates (bottom panel) change as schooling levels of the female population change. In the panels on the left, we consider changes in the proportion of individuals with no schooling (positive values imply more individuals moving into schooling), while the panels on the right consider changes in the share of individuals with primary schooling. The left hand panels are the conceptually cleaner transition to consider because positive movements in the variable always imply educational expansions, while in the case of primary school (in theory at least), positive movements *could* also imply movements from

secondary to primary (though in practice nearly all transitions are indeed positive).

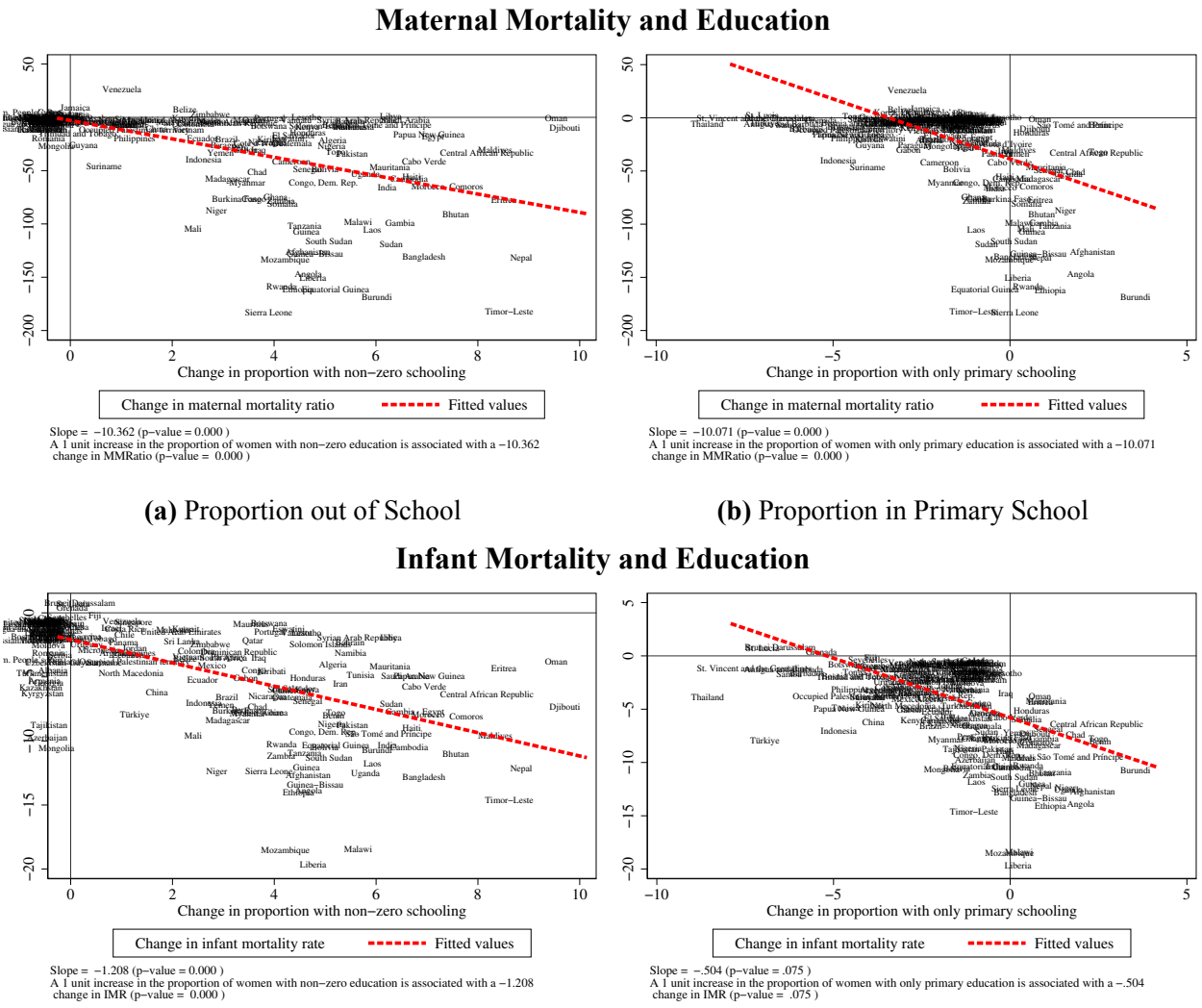
In each of the four panels, we observe a strong and highly significant ($p<0.01$) association between more schooling for women and lower maternal and infant mortality. Across 1990-2020, a 1 percentage point (pp) reduction in the proportion of women out of school is associated with 10.4 fewer maternal deaths per 100,000 live births, while a 1 pp increase in the proportion of women in primary school is associated with 10 fewer maternal deaths per 100,000 live births. A 1 pp decline in the proportion of women out of school is associated with 0.98 fewer infant deaths per 1,000 live births, and a 1pp increase in the share of women completing primary education lowers infant mortality by 1.12 in 1000 births. Both effects are large in magnitude. At baseline (1990) the maternal mortality ratio was 312 per 100,000 live births, and the infant mortality ratio was 50 per 100,000 live births. Thus, a 1 pp decline in women out of school is associated with a 3.2% decline in maternal mortality ratios, and a 2% decline in infant mortality rates. In Online Appendix Figure A3, we provide similar plots using residuals that purge impacts of changes in GDP (over the same time span) on education and mortality. The estimates are similar, which confirms that the associations hold independent of income or, put differently, improvements in women's education have broadly similar impacts in reducing mortality at a given level of income.

This strong relationship is *within* country over time (and holds whether or not we condition on economic growth rates). We also observe a strong *across* country relationship between education and mortality, see Figure 3. Here, each point is a country observation. The overlaid arrows indicate the direction of the within country correlation. Here we use the average years of women's education and mortality is in logarithms.[7] The arrows show the within-country associations over time, similar to Figure 2, but now country-specific. In the case of maternal mortality, only 17 of 185 countries have a positive correlation, and in the case of infant mortality, 6. This tells us that, in countries where women's education was growing, maternal and infant mortality were declining. The points under each arrow represent the country-level rates (not changes). The dots show a clear negative association, which is the cross-country relationship. This tells us that countries with higher levels of women's schooling have lower mortality rates.

The descriptive associations are compelling. However, they do not tell us whether policymakers that raise women's education would find that mortality had fallen. This would only be the case if the

[7]Mortality rates are bounded from below (at zero), and it is easier to produce larger declines in mortality when starting with a higher baseline level (low hanging fruit, diminishing returns to interventions). The relationship of education and mortality is hence more linear in proportions than in levels and taking logarithms accounts for this.

Figure 2: Maternal and Infant Mortality and Women's Education: Within country association of changes over time, 1990-2020



(a) Proportion out of School

(b) Proportion in Primary School

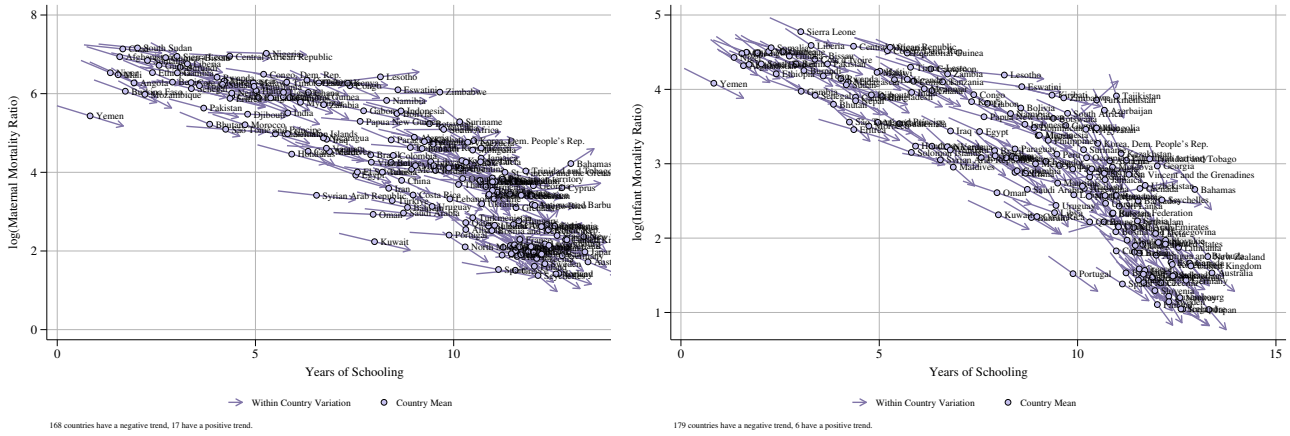
(c) Proportion out of School

(d) Proportion in Primary School

Notes to Figure 2: Each point represents the country average of Δ MMR (top panel) and Δ IMR (bottom panel) plotted against a Δ educational indicator. Δ is the first difference, and is a country average over the period 1990-2020. The inverse of proportion out of school is used, so Δ is interpreted as a *reduction* in the proportion of women out of school. Similar results are found when conditioning on changes in GDP per capita (see Online Appendix Figure A3).

increase in education had a causal effect on mortality. In a country-specific setting, the researcher would look for a policy experiment on education that created a large and sharp change in women's education and then trace impacts of this on the mortality outcomes. Since there is no universal policy experiment, it is challenging to identify quasi-experimental effects when looking at this relationship using a global database of countries, as we do in this paper. For the reasons discussed in the preceding section, panel data identification approximates causal effects. In the following subsections we discuss panel data estimates.

Figure 3: Maternal and Infant Mortality and Women's Education: Between and within country correlations, 1990-2020



(a) Maternal Mortality

(b) Infant Mortality

Notes to Figure 3: Points present country means for the full study period (1990-2020) of log mortality and female years of schooling. Arrows overlaid on each point present within country correlations between variables.

4.1.1 Maternal mortality- panel data estimates

Table 2 presents results for our principal specification (1), estimated on cross-country panel data for 185 countries observed at five yearly intervals between 1990-2020 resulting in a maximum of 185×7=1,295 observations. The dependent variable is the maternal mortality ratio in each country by year cell. This is regressed on a series of variables measuring the percentage of women in the particular country and year who have attained primary education and not further, secondary education and not further, and tertiary education, respectively. The omitted group is the percent of the population with no formal education. As discussed in the previous section, the identifying variation is within countries over time, as country fixed effects absorb all country-specific time-

invariant drivers of the outcomes, for instance, political institutions, climate, culture and deep-set attitudes towards women. In a revised specification, we add controls for country and time-varying variables that are potentially endogenous (ie driven by women's education and correlated with maternal mortality). If the coefficient on education diminishes when the controls are added, this is consistent with the controls being mechanisms for the education-mortality relationship.[8]

See column 1, where the only controls are country and year fixed effects.[9] We see a strong negative impact of women's education on maternal mortality. This is consistent with educated women being more successful at averting maternal death because they are more likely to adopt simple and low-cost practices to maintain hygiene, to recognize and react to symptoms such as bleeding or high blood pressure, more likely to assess information on the risks associated with abortion, or to fend off outside pressure to undertake abortion, more willing to seek professional birth attendance and likely to have fewer children. The effects are non-linear, showing high returns to some basic education and diminishing albeit positive returns to increasing the level of education. Moving one additional percentage of women into primary level schooling is associated with about 12 fewer maternal deaths (per 100,000 live births). With the share of women with primary (and tertiary) education held constant, moving one additional percent of women into having completed secondary schooling is associated with 8 fewer maternal deaths. These changes correspond to reduction of about 4% and 2.5% of the baseline average of maternal mortality ratios worldwide.

It is notable that there is only a small and statistically insignificant impact on mortality of increasing the share of women with tertiary education. This may reflect that the mechanism linking women's education to mortality outcomes for themselves and their children is the accrual (and diffusion) of basic health knowledge at low levels of education, which appears to be sufficient to influence health-seeking behaviours in and around childbirth. As these results draw from aggregate data, they will include the impacts of any spillovers from women with (at least) primary education to women with no education. Such spillovers are plausibly increasing in the proximity in which women live and the strength of community ties, factors which encourage information sharing and peer impacts on behavioural choices.

[8] In Appendix Table A1 we present models where instead of being measured as deaths per 100,000 live births, maternal mortality is measured as lifetime risk. Results are qualitatively similar.

[9] The largest potential sample contains 1,295 country-year cells. Here we have 1,238 country by year observations as a small number of countries do not have information for either maternal mortality or education.

Table 2: Maternal Mortality - The role of female educational attainment

	Maternal Mortality (per 100,000 live births)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Primary education (% Female)	-12.23***	-11.74***	-12.63***	-11.50***	-10.12***	-8.135***	-7.197***
	(1.790)	(1.863)	(2.234)	(2.127)	(2.422)	(2.428)	(2.413)
Secondary education (% Female)	-8.098***	-8.231***	-7.126***	-6.347***	-4.735**	-2.678	-2.319
	(1.164)	(1.202)	(1.233)	(1.177)	(1.818)	(2.006)	(1.945)
Tertiary education (% Female)	-2.900	-4.274	-1.701	-0.648	0.459	1.325	1.627
	(3.689)	(3.591)	(3.725)	(3.466)	(3.557)	(3.360)	(3.389)
Log GDPpc (PPP, 1000 USD)			-53.86**	-50.60**	-49.26**	-53.62**	-51.41**
			(25.38)	(22.90)	(20.01)	(20.66)	(19.79)
Immunization (DPT)				-1.551***	-1.302***	-1.335***	-1.378***
				(0.520)	(0.491)	(0.474)	(0.456)
Attended births					-1.266	-1.237	-1.098
					(0.955)	(0.975)	(0.963)
Fertility						47.33*	37.98
						(24.62)	(24.12)
Teen fertility							0.607*
							(0.339)
Observations	1238	936	936	936	936	936	936
MMR mean in 1990	312.1	275.8	275.8	275.8	275.8	275.8	275.8
R-squared	0.955	0.964	0.966	0.969	0.970	0.971	0.972
Number of countries	185	174	174	174	174	174	174

Notes: All regressions include fixed effects by country and year. Results are for the percent of the female population between the ages of 15 and 39 with each level of education indicated in variable names. The omitted baseline reference category is the population of women with no education. A full description of control variables is available in section 3. Standard errors clustered at the level of the country are displayed. * p<0.1; ** p<0.05; *** p<0.01.

Conditional estimates Columns 3-8 show estimates conditional upon log GDP, fertility, teen fertility and two proximate determinants of MMR – the share of the (adult female) population that gives birth with an attendant professional (roughly the inverse of home birth), and (child DPT) immunization rates. We include log GDP to allow that "income buys health and survival". This measure of economic growth will tend to capture impacts of other aspects of economic development that co-move with growth, including the development of legal and other institutions. The measure of maternal mortality that we use is per live birth and it is thus mechanically a function of the number of live births in the country. We include fertility as a control to allow that the normalized measure is, plausibly, causally dependent upon fertility (the number of births per woman). Women in developing countries have on average many more pregnancies than women in developed countries, and this is one reason that their lifetime risk of death due to pregnancy is higher. A woman's lifetime risk of maternal death—the probability that a 15 year old woman will eventually die from a maternal cause—is 1 in 3,800 in developed countries, versus 1 in 150 in developing countries. Given that teenage women face higher risks of complications and death as a result of pregnancy (Conde-Agudelo et al., 2005; Patton et al., 2009),[10] we also include the teen fertility rate. We include immunization rates to proxy for the state of public health coverage in a country (as is commonly done, for instance, by UNICEF).[11]

Publicly available data for these controls contain missing values for some country-year cells- the number of countries drops from 185 to 174 once we add controls to the model. Column 2 therefore provides results for the same specification as in column 1 (ie without the country-time varying controls), but now estimated on the data sample of country-year pairs for which all controls are observed. Column 2 thus acts as a bridge between columns 1 and 3. We might regard column 3 as the favoured specification, conditioning only on the log of GDP per capita. The coefficients are strikingly similar– changes in GDP do have a large mortality-reducing effect, but changes in education have a largely independent mortality-reducing effect. For instance, an additional one percent of the population having achieved primary education correlates with 12.6 fewer maternal deaths, which is 0.03 of a standard deviation and 4% of the mean maternal mortality rate in the sample. It is illustrative to consider the impact of moving the entire population of individuals with

[10] This may be because there is negative selection into early fertility, or because of an age-gradient in biological risk.

[11] Most immunization is directed at children under the age of 12 months (and sometimes up to 24 months). Our measure of immunization rates pertains to children. This measure however includes tetanus, and tetanus injections for pregnant mothers are part of routine prenatal care. In this way the immunization rates data we use may also directly influence maternal mortality.

no education in 2020—which on average is 14% of women—into primary education completion. Our estimates suggest this would lower MMR by 176 deaths, which is 0.43 of a s.d. of total MMR at baseline.

The addition of controls for immunization, attended births and fertility lowers the coefficient on primary education from 12.6 to a minimum of 7.2 fewer deaths, and it is consistently significantly distinct from zero (and not significantly different from the coefficient in column 2). The coefficient on secondary education remains negative but becomes small and statistically insignificant once all controls are in. An additional percentage of the female population moving into secondary education is associated with declines in maternal mortality of between 7.1 and 2.3 deaths (columns 2 to 7).

Impacts of income, fertility and public health coverage on mortality A 1% increase in GDP is associated with 51 fewer maternal deaths, which is 16 percent of the baseline mortality rate. A 1 SD increase in the log of GDP (1.17 at baseline) would thus map into approximately 60 fewer maternal deaths, or 19.2% of the baseline mean. A 1 pp increase in the immunization rate is associated with a reduction in maternal mortality of 1.38 deaths. A one standard deviation increase in immunization rates (which is 19.3% points) lowers maternal mortality by 26.6 deaths , which is 0.07 s.d. or 8.5% of the baseline mean. A 1 pp increase in the share of births attended by a skilled professional is correlated with approximately one less maternal death. A 1 s.d. increase (26.1% points) in the percentage of births attended by a skilled professional results in a drop in MMR of 28.6: an effect size similar to the immunization measure described above. A marginal increase in fertility – an additional birth per woman over her fertile life – is associated with an increase in maternal mortality of 38. A one s.d. increase in fertility (which is 1.91 births) leads to 72.6 additional maternal deaths. We also see a positive association of the teen fertility rate with MMR. One additional birth per 1,000 women aged 15-19 is associated with an increase of 0.6 maternal deaths. A one s.d. increase in births per 1,000 teens (which is 50.3 births) results in 30 more maternal deaths per 100,000 births, which is 0.07 s.d. and 9.8% of the mean.

Comparison of standardized effects of women's education on mortality with the effects of income, fertility and public health coverage In comparing standardised effects of education and GDP, these results suggest that a one standard deviation increase in the percent of women with primary education has larger effects on rates of maternal mortality ($-12.63 \times 12.78 = -161$) than a one standard deviation increase in the log of GDP per capita ($-53.86 \times 1.17 = -63.0$). This is

important, given that recent WHO factsheets on maternal mortality highlight differences in maternal mortality ratios by income and not education level (WHO, 2012). Another widely recognised stylized fact highlighted in the WHO factsheet is that maternal mortality is a function of health services. Again, if we use values from column (4), the standardised effect of primary education ($-11.50 \times 12.78 = -147$) is considerably larger than the standardised effect of our proxy of health service capacity ($-1.55 \times 19.32 = -29.96$). As laid out previously, the standardised impacts of fertility and teenage fertility are both also substantially lower than the standardised effect of primary education. To the extent that standardized estimates are useful, this suggests that the payoff to basic education in terms of a health sector outcome is equal to or larger than the payoff to a common index of provision in the health sector. This contributes to a literature documenting synergies between investments in education and health. It is of particular importance for policy given that WHO policy documents on maternal mortality have under-emphasised education.

4.1.2 Infant mortality- panel data estimates

In Table 3 we present estimates for infant mortality rates. The specifications are similar to those discussed for MMR and hence the discussion here is more concise.

As was the case for maternal survival, we see that women's education improves infant survival. We highlight some differences. For infant mortality, the gains from secondary education are similar (or a bit larger, rather than smaller) to the gains from primary education, though, as for MMR, there are no significant gains from expanding the share in tertiary education. Column 1 shows that for each additional percentage of the population moving into primary (or secondary) education, infant mortality declines by 0.8 per 1,000 live births, which is approximately 1.6% of the baseline mean. We can extrapolate these effects (all else held constant) to consider the impact of moving the entire share of women who have zero years of schooling into having completed primary school. Our estimates suggest this would generate a decline in infant mortality of -0.8$\times$14.4=-11.52, or approximately a 23% decline.

Conditional estimates These effects are highly stable to the inclusion of the time-varying controls, even though log GDP and immunization rates have a significantly negative impact on infant mortality. In contrast to the case for MMR, we see no significant association of skilled birth attendance or fertility (including teen fertility) with IMR.

Table 3: Infant Mortality - The role of female educational attainment

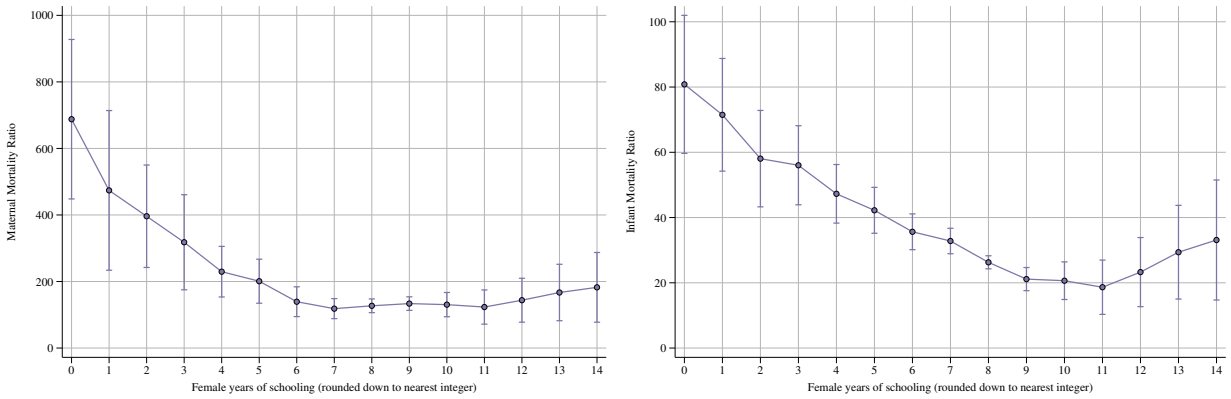
	Infant Mortality (per 1,000 live births)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Primary education (% Female)	-0.814***	-0.766***	-0.940***	-0.830***	-0.799***	-0.803***	-0.832***
	(0.201)	(0.193)	(0.127)	(0.119)	(0.140)	(0.164)	(0.191)
Secondary education (% Female)	-1.120***	-1.131***	-0.915***	-0.839***	-0.804***	-0.808***	-0.819***
	(0.124)	(0.135)	(0.130)	(0.121)	(0.139)	(0.138)	(0.147)
Tertiary education (% Female)	-0.605	-0.745*	-0.244	-0.141	-0.116	-0.118	-0.127
	(0.377)	(0.390)	(0.329)	(0.304)	(0.311)	(0.302)	(0.302)
Log GDPpc (PPP, 1000 USD)			-10.50***	-10.18***	-10.15***	-10.14***	-10.21***
			(1.289)	(1.142)	(1.127)	(1.097)	(1.169)
Immunization (DPT)				-0.152***	-0.147***	-0.146***	-0.145***
				(0.0470)	(0.0480)	(0.0481)	(0.0481)
Attended births					-0.0276	-0.0277	-0.0319
					(0.0421)	(0.0419)	(0.0426)
Fertility						-0.0925	0.192
						(1.728)	(1.826)
Teen fertility							-0.0185
							(0.0361)
Observations	1231	936	936	936	936	936	936
IMR mean in 1990	50.03	48.49	48.49	48.49	48.49	48.49	48.49
R-squared	0.968	0.973	0.981	0.983	0.983	0.983	0.983
Number of countries	184	174	174	174	174	174	174

Notes: All regressions include fixed effects by country and year. Results are for the percent of the female population between the ages of 15 and 39 with each level of education indicated in variable names. The omitted baseline reference category is the population of women with no education. A full description of control variables is available in section 3. Standard errors clustered at the level of the country are displayed. * p<0.1; ** p<0.05; *** p<0.01.

4.2 Marginal effects of education across the distribution of years of education

Using the panel data specifications in columns 3 of the preceding tables, we examine heterogeneity in the marginal impact of education by years of education, a more continuous measure than the categorical measures used thus far. The estimates are plotted in Figure 4. For both MMR and IMR, we observe diminishing returns to an additional year of education as the base level of education rises. In the case of maternal mortality (panel (a)), sharp declines are observed when the mean educational attainment of the population expands from 0-1 years, after which the marginal impactdeclines progressively and becomes essentially zero from around 6 years of education onwards (corresponding to primary in most countries). In the case of infant mortality, a nearly linear decline is observed up to around 9 years of education, which includes primary and secondary schooling.

Figure 4: Marginal Effects of Education on Maternal and Infant Mortality



(a) Maternal Mortality **(b)** Infant Mortality

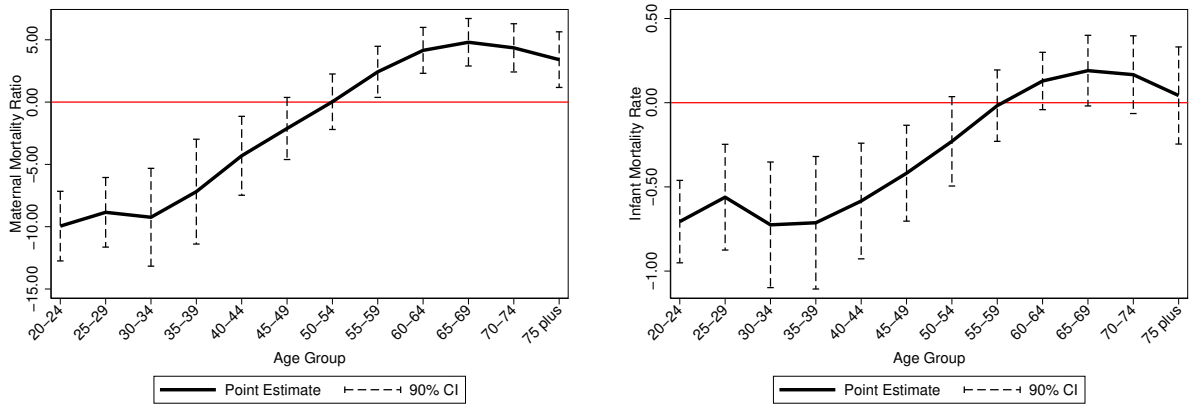
Notes to Figure 4: Estimates are plotted where (1) is estimated, but instead of a categorical indicators for the percent of women in each country×year with a particular education level, a series of binary indicators for mean education level (by year) for women in a country are included. Point estimates on each education level along with their 95% confidence intervals are plotted. Each specification additionally includes country by year fixed effects, and controls for GDP per capita.

4.3 Age profile of estimates

If education reduces maternal mortality risk mainly through increasing awareness and knowledge of the individual woman, we might expect that it is the educational attainment of fertile-aged

women that matters. However, if women's education also affects maternal mortality through "external" channels such as having more educated female health professionals, or through women's increased participation in policy-making, then we may find that female education at all ages, including post-fertile age matters.

Figure 5: Effect of Education on Mortality by Women's Age



(a) Maternal Mortality and Education by Age

(b) Infant Mortality and Education by Age

Notes to figure 5: Each point estimate and 95% confidence interval represent the results from a regression of country mortality (MMR or IMR in panel (a) and (b) respectively) on the proportion of women of indicated ages with completed primary education (as per specification 1). Each regression includes country and year fixed effects, as well as controls for GDP per capita. Standard errors are clustered by country.

The Wittgenstein education data record educational attainment by population age groups, from 15-19 year olds up to 70-74 olds, and age 75 plus. We replicate the baseline regression 1 which, earlier was run for fertile-aged women, using women of all ages and estimating 1 for each quinquennial age group from 20-24 up to 70-74, as well as the 75 plus group. We omit women age 15-19 from this analysis given that they are potentially in secondary education.

The age-specific estimates are plotted in figure 5. For parsimony we plot the coefficient on the proportion of primary educated women. It is overwhelmingly education at fertile ages (rather than all ages) that drives MMR reduction. Moreover, there is a sharp age-gradient in effects. Among 20-24 year olds, each additional percent of women with primary education results in a statistically significant reduction of around 10 maternal deaths per 100,000 live births, and this figure drops to less than 5 deaths per 100,000 live births for women aged 45-49, becoming essentially zero for ages over 55.[12] There is an age gradient in the same direction for infant mortality, the point

[12]We see small positive effects for education of women over the age of 65 which we do not interpret given the

estimates being close to -0.70 for women age 20-24, falling to close to -0.50 for women age 45-49 and becoming insignificantly different from zero after about age 60.

4.4 Regional estimates

The maternal mortality ratio in low income countries in 2020 was 430 per 100,000 births versus 12 per 100,000 in high income countries. There are large disparities between and within countries, with a few countries having extremely high maternal mortality ratios of 1,000 or more per 100,000 live births; more than half of all maternal deaths occur in sub-Saharan Africa (70% in 2020) while another 16% occur in South Asia.

As the base rates of maternal and infant mortality, and women's educational attainment vary substantially across the broad regions of the world, we re-estimated the panel data models by region using the specification in column (1), which conditions only on year and country fixed effects. See Tables 4 and 5. Results are similar when additionally conditioning on the log of GDP per capita, and a measure of health system capacity (see Appendix Tables A2 and A3), or if maternal mortality is measured as lifetime risk instead of being standardised by live births (Appendix Table A4). Although the margin at which women are gaining in education is different in different regions, a clear pattern across countries is that primary education matters—whether or not secondary and tertiary education also matter varies across regions.

Considering maternal mortality, we observe that expanding education for women has the largest impact in absolute terms in South and Central Asia, where the coefficients for primary and secondary education respectively indicate 38 and 8 fewer maternal deaths per 100,000 women and, where tertiary education is associated with a sizable impact even if this is not precisely determined. We see significant impacts in the other developing country regions, with some notable variation across regions in the importance of secondary relative to primary education. In Sub-Saharan Africa, a unit percentage increase in the share of women with primary education is associated with 14.6 fewer maternal deaths but secondary education has no impact. In the East Asia and Pacific region, declines in maternal death flow from increases in the share of women at every level of education-the declines associated with primary, secondary and tertiary education respectively are 10.6, 9.8 and 8.7. For the Latin America and Caribbean region we estimate reductions of 7.8 and 5.2 from primary and secondary education expansion respectively. In the Middle East and North Africa,

relatively small magnitudes and the fact that, in developing countries, there is selection into survival to age 65.

primary education has a smaller coefficient of 4.8 and secondary education is insignificant. In Europe and North America, improvements in the share with secondary and tertiary education show no association with mortality, and an increase in the share with primary education appears to increase mortality.[13]

Normalizing the effect sizes using baseline mortality rates (for 1990) only increases the variability of the estimates across regions. A 1pp increase in women in primary school corresponds to 8.6% of the baseline mean in South and Central Asia, 4.3-4.9% in Latin America and the Caribbean, East Asia and the Pacific, and the Middle East and North Africa, (with effects being larger in Asia and Latin America) and only 1.7% in Sub-Saharan Africa. The differences we observe across regions are plausibly related to different baseline levels of education and, amongst other things, to the extent to which education moves "mediator" or mechanism variables, discussed in the next section.

When considering infant mortality, effect sizes are slightly more muted in that educational expansions explain lower proportional declines in baseline infant mortality than those observed in maternal mortality. We nevertheless find largely similar geographic patterns. For instance, the largest estimates are again for South and Central Asia (where an additional percentage of women with primary and secondary education respectively results in 2 and 0.5 fewer deaths per 1,000 live births), and the East Asia and Pacific region again stands out because it is the one region where all levels of education, including tertiary, are mortality-reducing. When compared to baseline rates, the estimates range from -1.3% of the baseline mean for each pp expansion in primary education (Sub-Saharan Africa), up to -3.2% of the baseline mean (Latin America and the Caribbean). As in the case of maternal mortality, these relative or proportional effects are observed to largest in Latin America and in Asia, with these areas additionally being the only areas where effects are observed to persist beyond primary, and into secondary schooling.

[13] We do not put much weight on this as the share with primary- or zero- education is very small.

Table 4: Maternal Mortality and Female Education – Stratifying by Region

	(1) East Asia & Pacific	(2) Europe & North America	(3) Latin America & Caribbean	(4) Middle East & North Africa	(5) South & Central Asia	(6) Sub-Saharan Africa
Primary education (% Female)	-10.63***	1.542	-7.772**	-4.787*	-38.27***	-14.58**
	(2.807)	(1.072)	(3.520)	(2.624)	(8.202)	(5.567)
Secondary education (% Female)	-9.822***	1.507	-5.163**	-1.283	-8.320***	1.201
	(1.961)	(1.150)	(2.448)	(1.490)	(1.554)	(5.255)
Tertiary education (% Female)	-8.725**	0.611	-2.883	3.047	-16.63	24.54
	(3.159)	(1.091)	(6.516)	(1.846)	(13.52)	(26.06)
Observations	172	315	209	137	86	319
MMRatio mean in 1990	219.9	23.97	159.3	108.4	443.6	810.1
R-squared	0.962	0.615	0.879	0.871	0.974	0.907
Number of countries	26	45	32	21	13	48

Notes: All regressions include fixed effects by country and year. Results are for the percent of the female population between the ages of 15 and 39 with each level of education in each country. Standard errors clustered at the level of the country are displayed. * p<0.1; ** p<0.05; *** p<0.01

Table 5: Infant Mortality and Female Education – Stratifying by Region

	(1) East Asia & Pacific	(2) Europe & North America	(3) Latin America & Caribbean	(4) Middle East & North Africa	(5) South & Central Asia	(6) Sub-Saharan Africa
Primary education (% Female)	-1.169***	0.668*	-1.190**	-0.797***	-2.069**	-1.260***
	(0.405)	(0.336)	(0.542)	(0.243)	(0.828)	(0.386)
Secondary education (% Female)	-1.563***	-0.123	-1.232***	-0.346	-0.574**	-0.118
	(0.452)	(0.366)	(0.298)	(0.221)	(0.231)	(0.442)
Tertiary education (% Female)	-1.723*	0.220	-0.146	-0.521	-1.441	0.523
	(0.944)	(0.379)	(1.188)	(0.372)	(1.128)	(1.663)
Observations	172	315	202	137	86	319
IMR mean in 1990	42.32	16.08	37.41	34.96	77.36	96.05
R-squared	0.956	0.849	0.952	0.945	0.991	0.941
Number of countries	26	45	31	21	13	48

Notes: All regressions include fixed effects by country and year. Results are for the percent of the female population between the ages of 15 and 39 with each level of education in each country. Standard errors clustered at the level of the country are displayed. * p<0.1; ** p<0.05; *** p<0.01

4.5 Potential pathways and correlated effects

The results presented so far establish that gains in the share of educated women have globally been associated with declines in maternal mortality with the exception of Europe and North America, where basic education is almost universal and where we see no impact of improvements since 1990. We also saw that, conditional upon women's education, improvements in GDP and public health provision, and declines in fertility have further impacts on mortality.

.

Table 6: Education Correlates

	Age of marriage	Births Attended	Contracep Use	Control over Reproduc.	Fertility	Teen Fertility
	(1)	(2)	(3)	(4)	(5)	(6)
Primary education (% Female)	0.0167	1.091***	0.374***	-0.174	-0.0357***	-2.531**
	(0.0208)	(0.351)	(0.103)	(1.031)	(0.00722)	(1.086)
Secondary education (% Female)	-0.0473**	1.274***	0.434***	0.136	-0.0350***	-1.742*
	(0.0217)	(0.131)	(0.0546)	(0.696)	(0.00651)	(0.900)
Tertiary education (% Female)	0.0430	0.875***	-0.284	1.926	-0.0281	-1.663
	(0.0597)	(0.334)	(0.298)	(2.237)	(0.0175)	(1.669)
Log GDPpc (PPP, 1000 USD)	0.439*	1.059	2.625	-0.878	0.276**	-2.891
	(0.236)	(3.808)	(2.034)	(21.80)	(0.117)	(10.65)
Immunization (DPT)	0.00325	0.197**	-0.00406	-0.172	-0.00144	0.0563
	(0.00750)	(0.0767)	(0.0308)	(0.241)	(0.00195)	(0.139)
Contraception use					-0.0207***	-0.282
					(0.00602)	(0.196)
Observations	766	936	686	100	686	686
Mean in 1990	23.41	74.85	41.43	.	4.297	87.54
R-squared	0.947	0.954	0.968	0.982	0.984	0.936
Number of countries	168	174	161	64	161	161

Notes: All regressions include fixed effects by country and year. Results are for the percent of the female population between the ages of 15 and 39 with each level of education in each country. A full description of control variables is available in section 3. Standard errors clustered at the level of the country are displayed. * p<0.1; ** p<0.05; *** p<0.01.

In this section we examine the extent to which within-country improvements in women's education move some of the proximate determinants of MMR and IMR, as laid out in Section 3. Specifically, we consider fertility-related variables (age at marriage, control over reproductive behaviour, fertility and teen fertility), and indicators of reproductive health coverage (share of births

attended by a skilled professional) and the share of women using (modern) contraception, variation in which can reflect either or both of fertility preferences and reproductive health coverage.

See Table 6 where each of these mediators is regressed on the same educational measures, country and year fixed effects, and covariates. We see that increases in women's education are associated with a significant increase in the share of women using contraception and, consistent with this, with declines in fertility and teen fertility. Primary and secondary education shares have roughly similar impacts, tertiary has no impact. There is, somewhat unexpectedly, no discernible impact on the share of women reporting control over reproduction, though unlike other variables, the coverage of this measure is particularly low. Secondary, but not primary education, tends to lower women's age at marriage. The percentage of births attended by a skilled professional responds to increases in education at all levels, including tertiary, and the magnitudes are comparable across levels. Thus the studied variables are potential pathways- education not only directly influences maternal mortality, it is also correlated with other determinants of maternal mortality.

These correlates are potentially relevant in illuminating regional variation in observed returns to education. In Table 7 we present similar models as those in Table 6, however here separately by the same regions considered in Section 4.4. We do not include in this analysis the measure of women's control over reproduction given very low coverage when stratifying by region, and omit teen fertility given broadly similar patterns to total fertility. In Panel A, we observe that education correlates most strongly with birth attendance precisely in the regions where education is most strongly linked to maternal mortality declines (Asia and Latin America). We similarly observe that large and significant relationships with fertility exist in these regions where education and mortality links are strongest, and additionally these relationships exist with expansions of education both at primary and secondary level. While only descriptive, such patterns are able to rationalise the geographic divergence in impacts of education on mortality.

We can plug the relationships estimated in Table 6 into the specifications in Table 2. For instance, using column 7 in Table 2, we estimate a one s.d. reduction in fertility led to a 0.23 s.d. decline in MMR. Since Table 6 column 5 tells us that a single percentage increase in primary education years leads to a 0.008 s.d. decline in fertility, it follows that there is a fertility-mediated impact of primary education on MMR of 0.08×0.23 or 0.002 s.d. This is in addition to the direct impact of primary education on MMR in Table 2, which is 0.04 s.d. When considering that the total percent of women without formal education is much larger than the one percent which the preced-

Table 7: Education Correlates by Region

	(1) East Asia & Pacific	(2) Europe & N. America	(3) LAC	(4) MENA	(5) South & C. Asia	(6) SSA
Panel A: Birth Attendance						
Primary education (% Female)	0.699	-0.425	1.028***	0.656	1.459**	0.147
	(0.430)	(0.270)	(0.264)	(0.485)	(0.490)	(0.476)
Secondary education (% Female)	1.022***	0.0639	1.037***	0.757**	1.123***	0.338
	(0.351)	(0.101)	(0.233)	(0.305)	(0.229)	(0.514)
Tertiary education (% Female)	0.787	-0.256	0.572	-0.302	0.410	-0.663
	(0.487)	(0.162)	(0.687)	(0.460)	(0.918)	(1.385)
Observations	136	251	186	100	73	235
Mean in 1990	81.57	98.65	78.99	67.63	62.89	51.22
Panel B: Age at Marriage						
Primary education (% Female)	-0.0208	0.103	0.396***	-0.0366	0.0854	0.0379
	(0.0493)	(0.0846)	(0.0984)	(0.0638)	(0.0540)	(0.0544)
Secondary education (% Female)	-0.0445	0.0381	-0.00952	0.0361	0.0292*	0.0161
	(0.0574)	(0.0585)	(0.101)	(0.0428)	(0.0150)	(0.0682)
Tertiary education (% Female)	0.0396	0.147	0.505*	0.0763	0.0584	-0.128
	(0.120)	(0.130)	(0.248)	(0.0696)	(0.0629)	(0.114)
Observations	105	222	124	78	57	211
Mean in 1990	23.99	24.75	24.56	23.64	20.16	20.93
Panel C: Fertility						
Primary education (% Female)	-0.0424***	0.0168	-0.0459***	-0.0651**	-0.0786***	-0.0132
	(0.0116)	(0.0105)	(0.0138)	(0.0267)	(0.0251)	(0.0182)
Secondary education (% Female)	-0.0344***	-0.000662	-0.0490***	-0.0141	-0.0479***	-0.0248
	(0.00942)	(0.00648)	(0.0114)	(0.0136)	(0.0124)	(0.0242)
Tertiary education (% Female)	-0.0142	0.0206**	-0.0396	-0.00350	0.0264	0.0181
	(0.0174)	(0.00885)	(0.0300)	(0.0401)	(0.0579)	(0.0324)
Observations	172	315	209	137	86	319
Mean in 1990	3.708	1.935	3.508	4.731	4.817	6.169
Panel D: Contraceptive Use						
Primary education (% Female)	0.297	-0.613	1.048***	0.549**	1.448***	-0.501
	(0.558)	(1.908)	(0.218)	(0.227)	(0.380)	(0.326)
Secondary education (% Female)	0.171	-0.120	0.424	-0.0537	0.482	0.357
	(0.475)	(1.620)	(0.284)	(0.260)	(0.290)	(0.392)
Tertiary education (% Female)	-0.494	-1.083	0.751	-0.573	1.020	-1.977
	(0.758)	(1.688)	(0.644)	(0.354)	(1.628)	(1.258)
Observations	98	130	119	77	62	232
Mean in 1990	52.24	72.40	49.20	39.09	29.09	20.64

Notes: Refer to Notes to Table 5. Identical models are estimated, however here stratifying by region.

ing calculations assume, these effects can become quite large. Indeed, if we scale these effects by the 14.4% of women of fertile age with no formal schooling in 2020 according to Wittgenstein data, this would suggest direct effects of a 0.58% sd decline in MMR, plus an additional 0.03% sd decline owing to fertility-mediated effects. Similar calculations can be made for education-led impacts of birth attendance and teenage fertility. In the case of birth attendance, the education mediated effects of moving all uneducated women into primary education would bring about an additional 0.014% sd decline in maternal mortality, and in the case of teenage fertility, the mediated effect would be an additional 0.03% sd decline. Calculating the impacts of moving all women without education into (at least) primary school is a policy consideration. Overall, these results show that education is strongly positively correlated with fertility and birth attendance conditional upon country fixed effects and aggregate trends. While this is simply an accounting exercise, and as such the estimated associations will not be entirely causal, causal components such as the degree to which a more educated population demands higher public health services, and how this maps into mortality declines merit testing.

5 Conclusion

In the last three decades, maternal mortality has declined sharply (by 50%), infant mortality has seen continuing decline. Yet, every day in 2020, almost 800 women and 4 million infants died from preventable causes. Almost 95% of maternal and infant deaths occurred in low and lower middle-income countries. The analysis in this paper suggests that some part of these declines may be attributed to the sharp rise in women's education over the same period. Equally, some part of the continuing scourge of avoidable deaths is the result of a failure to universalize education for girls. The mechanisms by which the data indicates that women's education may act upon the mortality outcomes include a reduction in fertility and especially teen fertility and an increased tendency to give birth with a skilled professional present. Our analysis reveals that basic education has the power to deliver maternal and infant mortality reduction: the marginal impacts of education are largest at the lower end of the education distribution. They are largest among younger women. There is considerable variation in the estimated relationships across regions, with South and Central Asia exhibiting the strongest relationship. Rough calculations based on the experience of the last 30 years suggest that moving all women worldwide with no schooling to 1 year of education could account for reductions in MMR of as much as 180 per 100,000 live births, while moving women from 5 to 6 years could results in approximately 20-25 fewer deaths per 100,000 live births.

Our results suggest that it is not just general education rates in a society which matter (or education acting as a proxy for economic development). The results hold conditional upon GDP and reductions in MMR are nearly exclusively driven by educational attainment of fertile-age women. These findings are relevant for policy makers considering progress towards the sustainable development goals. The evidence in this paper suggests strong synergies between improvements in education, gender equality, and health, which are set out as separate development goals.

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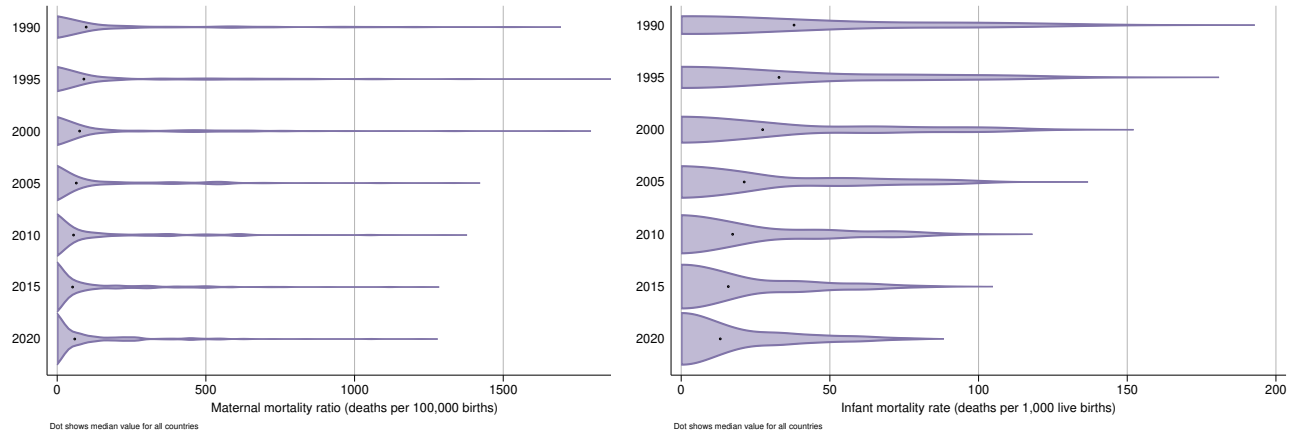
Online Appendices for:

"Education Expansion as a Policy Tool for Reductions in Infant and Maternal Mortality"

Sonia Bhalotra Damian Clarke

For online publication.

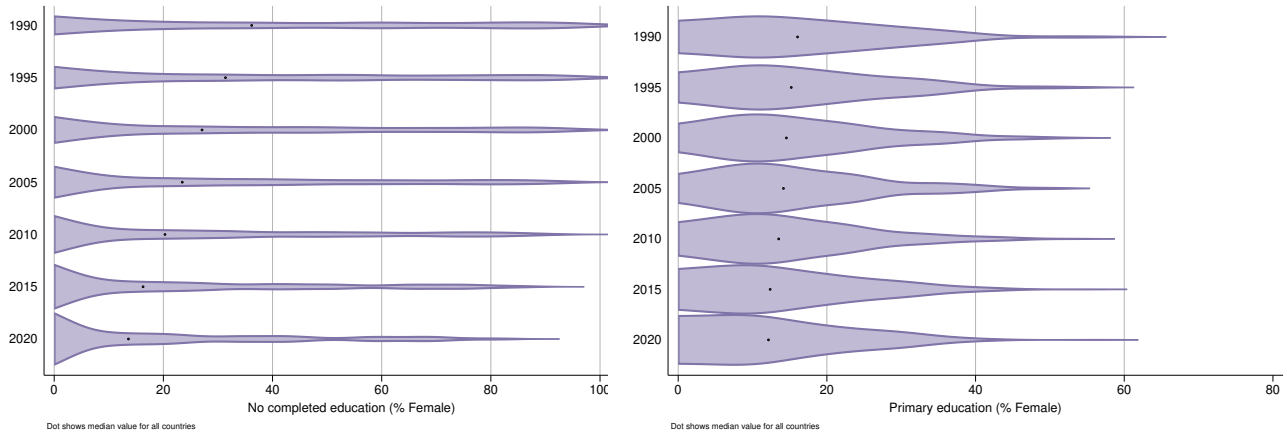
Figure A1: Time Trends: Maternal and Infant Mortality



(a) Maternal Mortality **(b)** Infant Mortality

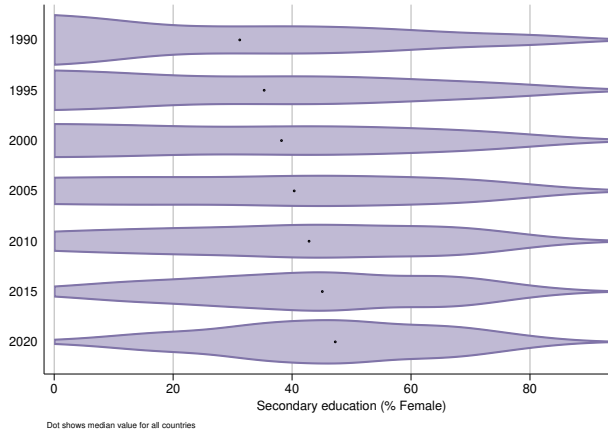
Notes to Figure A1: Plots present a distributional plot for mortality in each year (displayed as horizontal histograms), with median values at each period indicated as a solid point. For each year, thicker values of the horizontal histogram imply a greater mass of countries at these values.

Figure A2: Time Trends: Women's Education

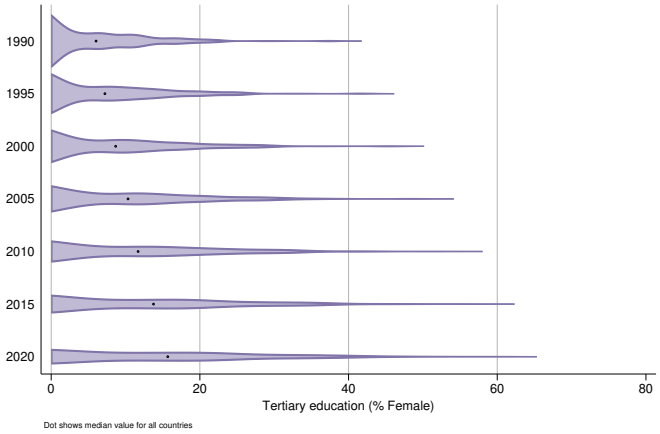


(a) No Completed Education

(b) Primary Education



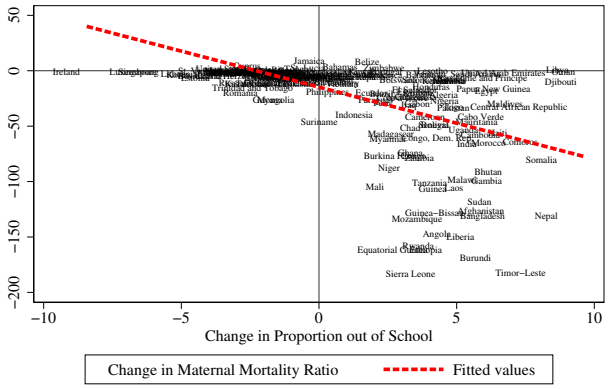
(c) Secondary Education



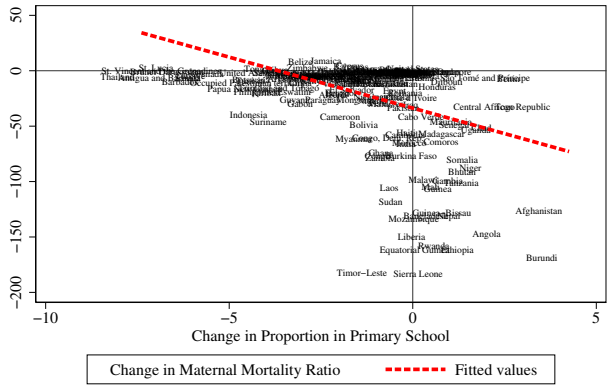
(d) Tertiary Education

Notes to Figure A2: Plots present a distributional plot for educational levels in each year (displayed as horizontal histograms), with median values at each period indicated as a solid point. For each year, thicker values of the horizontal histogram imply a greater mass of countries at these values.

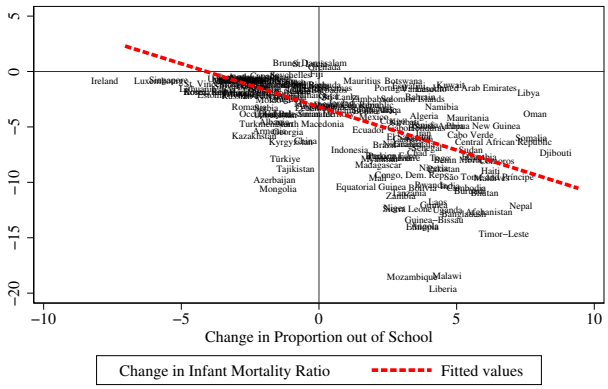
Figure A3: Maternal Mortality Ratio and Women's Education (Conditional on Income)



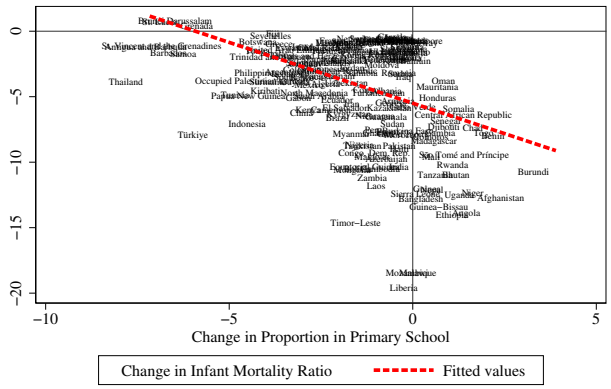
(a) Proportion out of School (MMR)



(b) Proportion in Primary School (MMR)



(c) Proportion out of School (IMR)



(d) Proportion in Primary School (IMR)

Notes to Figure A3: Each point represents the country average of Δ MMR (top panels) and Δ IMR (bottom panels) plotted against a Δ educational indicator after conditioning on Δ GDP per capita. Δ is the first difference, and is a country average over the period 1990-2020. The inverse of Proportion out of School is used, so Δ is interpreted as a *reduction* in the proportion of women out of school. Unconditional results are presented in Figure 2 of the paper.

Table A1: Lifetime Maternal Mortality Risk - the role of female educational attainment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Primary education (% Female)	-0.0674***	-0.0633***	-0.0664***	-0.0605***	-0.0625***	-0.0413***	-0.0452***
	(0.0121)	(0.0102)	(0.0127)	(0.0112)	(0.0115)	(0.0138)	(0.0156)
Secondary education (% Female)	-0.0340***	-0.0354***	-0.0315***	-0.0274***	-0.0297***	-0.00770	-0.00922
	(0.00868)	(0.00782)	(0.00836)	(0.00820)	(0.00930)	(0.0142)	(0.0144)
Tertiary education (% Female)	0.0229	0.00945	0.0185	0.0240	0.0225	0.0317	0.0304
	(0.0269)	(0.0237)	(0.0262)	(0.0252)	(0.0251)	(0.0235)	(0.0230)
Log GDPpc (PPP, 1000 USD)			-0.190	-0.173	-0.175	-0.222	-0.231
			(0.147)	(0.136)	(0.136)	(0.139)	(0.143)
Immunization (DPT)				-0.00808**	-0.00843**	-0.00878***	-0.00860***
				(0.00355)	(0.00330)	(0.00329)	(0.00330)
Attended births					0.00178	0.00210	0.00151
					(0.00531)	(0.00569)	(0.00579)
Fertility						0.505***	0.545***
						(0.192)	(0.195)
Teen fertility							-0.00257
							(0.00196)
Observations	1238	936	936	936	936	936	936
Lifetime risk mean in 1990	1.801	1.517	1.517	1.517	1.517	1.517	1.517
R-squared	0.938	0.953	0.954	0.956	0.956	0.961	0.961
Number of countries	185	174	174	174	174	174	174

Notes: All regressions include fixed effects by country and year. Results are for the percent of the female population between the ages of 15 and 39 with each level of education in each country. A full description of control variables is available in section 3. Standard errors clustered at the level of the country are displayed. * p<0.1; ** p<0.05; *** p<0.01

Table A2: MMRatio - the role of female educational attainment by region

	(1) East Asia & Pacific	(2) Europe & North America	(3) Latin America & Caribbean	(4) Middle East & North Africa	(5) South & Central Asia	(6) Sub-Saharan Africa
Primary education (% Female)	-12.07***	1.166	-7.228**	-4.833**	-27.61***	-12.84**
	(2.723)	(1.218)	(3.169)	(2.149)	(6.183)	(5.175)
Secondary education (% Female)	-12.35***	1.391	-2.771	-1.880**	-4.816**	0.868
	(1.962)	(1.118)	(2.194)	(0.812)	(1.662)	(5.607)
Tertiary education (% Female)	-12.68***	0.730	-0.0161	4.740	7.486	19.56
	(3.538)	(1.287)	(6.632)	(2.899)	(13.27)	(24.93)
Log GDPpc (PPP, 1000 USD)	37.85***	-20.13**	-43.35	-31.98	-204.0**	-236.1**
	(5.407)	(9.741)	(35.87)	(23.63)	(79.87)	(87.91)
Immunization (DPT) (%)	-0.988*	-0.211	-0.515	-1.829	-2.863	-2.322*
	(0.504)	(0.562)	(0.466)	(2.075)	(2.257)	(1.266)
Observations	162	300	188	119	83	301
MMRatio mean in 1990	159.2	24.43	170.3	106.6	370.9	781.5
R-squared	0.976	0.642	0.921	0.888	0.980	0.924
Number of countries	25	45	29	19	13	46

Notes: All regressions include fixed effects by country and year. For the full list of countries by year see online appendix table 1. Results are for the percent of the female population between the ages of 15 and 39 with each level of education in each country. Standard errors clustered at the level of the country are displayed.
* p<0.1; ** p<0.05; *** p<0.01

Table A3: Infant Mortality - the role of female educational attainment by region (with covariates)

	(1) East Asia & Pacific	(2) Europe & North America	(3) Latin America & Caribbean	(4) Middle East & North Africa	(5) South & Central Asia	(6) Sub-Saharan Africa
Primary education (% Female)	-0.712***	0.546*	-1.054**	-0.691***	-1.173	-1.161***
	(0.199)	(0.310)	(0.507)	(0.173)	(0.731)	(0.312)
Secondary education (% Female)	-1.065***	-0.154	-0.973**	-0.470***	-0.344*	-0.0999
	(0.205)	(0.325)	(0.377)	(0.131)	(0.191)	(0.465)
Tertiary education (% Female)	-0.988**	0.315	0.450	-0.473**	0.468	0.371
	(0.359)	(0.345)	(1.266)	(0.198)	(1.101)	(1.367)
Log GDPpc (PPP, 1000 USD)	-8.552***	-11.89***	-6.799	-9.135**	-16.18***	-15.42***
	(1.267)	(2.928)	(5.933)	(3.606)	(5.069)	(4.400)
Immunization (DPT) (%)	-0.277***	-0.0254	-0.0531	-0.186*	-0.187**	-0.0528
	(0.0672)	(0.0754)	(0.0549)	(0.0912)	(0.0814)	(0.0764)
Observations	162	300	188	119	83	301
IMR mean in 1990	36.46	16.27	38.92	34.43	74.46	92.94
R-squared	0.986	0.893	0.957	0.977	0.994	0.952
Number of countries	25	45	29	19	13	46

Notes: All regressions include fixed effects by country and year as well as covariates displayed in Table output. Standard errors clustered at the level of the country are displayed. * p<0.1; ** p<0.05; *** p<0.01

Table A4: Lifetime Maternal Mortality Risk - the role of female educational attainment by region

	(1) East Asia & Pacific	(2) Europe & North America	(3) Latin America & Caribbean	(4) Middle East & North Africa	(5) South & Central Asia	(6) Sub-Saharan Africa
Primary education (% Female)	-0.0545**	0.00369*	-0.0515***	-0.0212	-0.295***	-0.107**
	(0.0201)	(0.00187)	(0.0175)	(0.0126)	(0.0272)	(0.0424)
Secondary education (% Female)	-0.0447***	0.00237	-0.0298***	0.00487	-0.00867	0.0101
	(0.0131)	(0.00207)	(0.00995)	(0.00997)	(0.0104)	(0.0300)
Tertiary education (% Female)	-0.0373*	0.00225	-0.0273	0.0366**	-0.0395	0.130
	(0.0196)	(0.00195)	(0.0260)	(0.0164)	(0.0354)	(0.146)
Observations	172	315	209	137	86	319
Lifetime risk mean in 1990	1.049	0.0496	0.642	0.582	2.443	5.117
R-squared	0.898	0.615	0.893	0.853	0.981	0.913
Number of countries	26	45	32	21	13	48

Notes: All regressions include fixed effects by country and year. Results are for the percent of the female population between the ages of 15 and 39 with each level of education in each country. Standard errors clustered at the level of the country are displayed. * p<0.1; ** p<0.05; *** p<0.01

A Maternal mortality in the medical literature

Between 1990 and 2010, maternal mortality worldwide dropped by almost 50%. While this is impressive, certainly relative to trends in the preceding twenty years, the average rate of decline in the global maternal mortality ratio (the number of maternal deaths per 100,000 live births) of 3.1% per year over this period falls well below the annual decline of 5.5% required to achieve the Millennium Development Goals (MDG) for maternal health adopted by the international community in 2000. There are signs of increasing policy commitment to addressing maternal mortality: official development assistance for maternal and new-born health has risen relative, for instance, to funding for child health (Greco et al., 2008).[14]

The medical literature suggests that maternal mortality is principally associated with labour, delivery and the early postpartum period; largely occuring between the third trimester of pregnancy and the first post-partum week. The principal medical cause of these deaths is obstetric haemorrhage, a largely preventable cause provided "access to timely and competent obstetric care" is available (Ronsmans and Graham, 2006). Along with haemorrhage, a number of other causes are reported, including sepsis, abortion, hypertensive disorders, obstructed labour, embolism and ectopic pregnancy[15]. Despite the fact that these complications account for the majority of deaths during labour, birth and the post-partum period, the prevalence of each varies by region (Khan et al., 2006). Whilst haemorrhage is the principal cause in developing regions, complications from caesarian sections and anaesthesia are more common in developed countries, and other region-specific disease burdens are noted (abortion in Latin America and the Caribbean, sepsis and HIV in Africa and anaemia in Asia).

Much focus in the existing literature is on the difficulty of obtaining accurate statistics of maternal mortality, particularly in those areas where maternal mortality is highest(Ronsmans and Graham, 2006; McCarthy and Maine, 1992; McAlister and Baskett, 2006). The ability to compile

[14] Overall donor disbursements increased from US$2,119 million in 2003 to $3,482 million in 2006; funding for child health increased by 63% and that for maternal and new-born health increased by 66%. In the 68 priority countries, child-related disbursements increased from a mean of $4 per child in 2003 to $7 per child in 2006; disbursements for maternal and neonatal health increased from $7 per live birth in 2003 to $12 per live birth in 2006.

[15] Highly cited figures (AbouZahr et al., 1991) suggest figures of 25% due to haemorrhage, 20% due to indirect causes, 15% due to infection, 13% due to abortions, 12% due to eclampsia, 8% to obstructed labour and 8% to other direct causes. Khan et al. (2006) provide evidence broadly in agreement with these values however note that these estimates suffer from large confidence intervals.

credible data is confounded by the lack of consistent classification prior to the introduction of ICD-9 coding, the need to collect data from an array of data sources, misclassification of maternal death, significant under-reporting of maternal mortality, and missing and incomplete data(Yazbeck, 2012; Hogan et al., 2010).

Early work[16] in the medical literature highlighted the importance of addressing the 'proximal causes' or inputs of maternal mortality; namely the likelihood that a women becomes pregnant, or the likelihood of complications (and the treatment of these complications) conditional upon her becoming pregnant (McCarthy and Maine, 1992; Goodburn and Campbell, 2001; Trussell and Pebley, 1984). Along with attendence and medical care for mothers during pregnancy and childbirth, these studies suggest that altering maternal age, quantity of births, and spacing of births could have a considerable effect on rates of maternal mortality.

More recent medical studies also suggest that socioeconomic factors are fundamental in the reduction of maternal mortality. Costello et al. (2004), for example, argue that the focus on primary care and skilled attendance is not a sufficient strategy, citing considerable reductions in maternal mortality following community-based interventions[17]. McAlister and Baskett (2006) suggest that social—and specifically gender-specific factors such as female education—should predict maternal mortality, demonstrating cross-country correlation between these variables. Recent work from Ahmed et al. (2012) suggests that family planning inputs would allow for considerable reductions in maternal deaths world-wide, with the authors estimating that the equivalent of 44% of maternal deaths in 2008 were averted due to contraceptive use, and that meeting unmet demand for family planning could prevent a further 29%. Finally, the importance of socio-economic factors is also highlighted by Ahmed et al. (2010) (and earlier by Shen and Williamson (1999)) who suggest that more educated and empowered women are more likely to utilise health services related to maternal mortality such as antenatal care, and skilled birth attendants, and by Bhutta and Black (2013) who highlight the social determinants of maternal health, and the link of poor health to poorly served urban slum environments where social and educational support networks are weak.

[16]The Safe Motherhood Initiative of 1987 was put in place in response to the increasing recognition of this topic in developing countries.

[17]Also see Manandhar et al. (2004) for the results of a randomized trial of community-based meetings.